E-R MODEL

★ E-R Model means Entity Relationship Model.
- ↳ Used for designing databases.
- ↳ Defines (Logical) conceptual view of a database.

★ **Entity** (Thing or Object that is present)

↳ It can be anything that has an independent existence & about which we collect data.

Represented by Rectangles

→ Become Table in relational model.
→ Have attributes that give them their identity.

Entity Set - Collection of similar types of entities, that shares same attributes.

★ **Attributes** (Property / Characteristics of an Object)

↳ For each attribute there is a set of permitted values, called domain (or range) of that attribute.

Represented by Ellipse

Attribute Types -
- Simple & Composite
  - Simple → Atomic Values (Cannot Further Divide) Ex. Birth Date
  - Composite → (Further Dividable) Ex. Name
- Single-Valued & Multi-Valued
- Stored & Derived
  - Derived → from other attributes in database, Ex - age.

★ **Relationship** - An association among entities.

Represented by diamond-shaped box.

Relationship Set - A set of relationships of similar type.

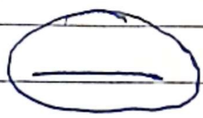
* Descriptive Attribute - Attributes of a relationship.

* The no. of different entity sets participating in a relationship set are called as degree of a relationship set.

Unary, Binary, Ternary, n-ary Relationships.

* Multi-Valued Attributes are represented by Double Ellipses.

* Key-Attribute - The attribute which uniquely identifies each entity in the entity set is called key attribute.
  - It represents a primary key.
  - Represented by an ellipse with underlying lines.



* Cardinality defines the no. of entity of an entity set that participates in a relationship set.

  Of 4 Types -
  1) One-to-One
  2) One-to-Many
  3) Many-to-One
  4) Many-to-Many

* For 1 ⟶ , For Many ——

* Total Participation - Each entity is involved in the relationship. (Represented by double lines)

* Partial Participation - Not all entities are involved in the relationship. (represented by single line)

★ A key is an attribute or set of attributes that uniquely identifies any record (or tuple) from the table.

Types of keys -

1) Superkey - A combination of all possible attributes that can uniquely identify the rows (or tuples) in the given relation.

Superset of candidate key.

Primary Key   Candidate Key   Alternate key

Super key

May contain extra/unnecessary attributes

2) Candidate key - Minimal Superkey

No extra attribute can be removed

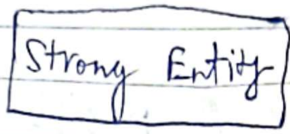
3) Primary key -

4) ~~Candidates~~ Alternate key - Candidate keys that were not chosen as primary key.

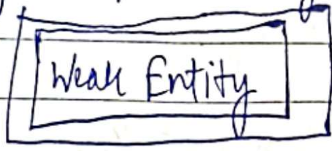
5) FOREIGN KEY - A key used to link 2 tables.
To ensure referential integrity of the data.

6) Composite key - A key that has more than one attributes is known as composite key. It is also known as Compound key.

☆ Strong entity always have a primary key (independent of any other entity.)
Represented By Single Rectangle.



☆ Weak entity doesn't have sufficient attributes to form a Primary key. Represented by double Rectangle.



☆ EER Model (Extended ER Model)

opposites

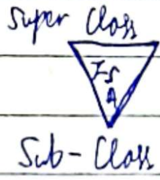
imp. feature of both

Generalization

- Process of extracting common properties from a set of entities & create a generalized entity from it.
- Bottom-Up approach
- Concept of Superclass & Sub-Class

Specialization

- Top-Down approach
- An entity is broken down into sub-entities based on their characteristics.



Inheritance

- It allows lower level entities to inherit the attributes of higher level entities.